Analytic Neutron Wall Loading for Spin-Polarized Fusion in Quasi-Axisymmetric Stellarators

Field-period perturbation of the axisymmetric kernel

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Where this started: four papers

This project grew out of reading a quartet:

- **Kulsrud** (1986) — *Physics of spin-polarized plasmas.* Polarizing D and T spins enhances the DT reactivity ($\sim$50%) and makes the fusion emission *anisotropic* about $\hat{\boldsymbol{B}}$.
- **Schwartz** (2025) — *Analytic neutron wall loading from spin-polarized fusion in axisymmetric geometries.* Closed-form, differentiable NWL for tokamaks via incomplete elliptic integrals F, E.
- **Bae et al.** (2025) — *Neutronics of spin-polarized fuel in spherical tokamaks.* Monte-Carlo evidence the steering matters at the wall / center stack.
- **Lion et al.** (2022) — *Deterministic 3D neutron wall load for generic stellarators.* Fast, but integrates the toroidal direction *numerically*.

The gap: Schwartz's closed form is exact only in axisymmetry; Lion is 3D but numerical. *Can we keep Schwartz's closed, differentiable form in a stellarator?*

The object: neutron wall load (free-streaming)

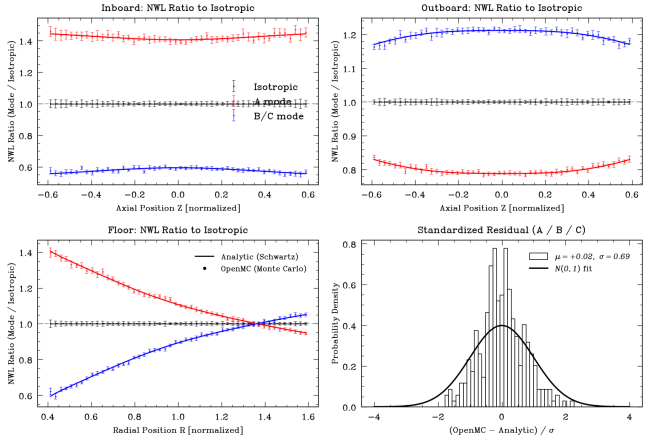
First-flight neutron current through a wall patch, weighted by the polarized emission kernel:

$$Q_{\mathrm{NWL}}(\boldsymbol{r}_w) \;\propto\; \int_{V_s} \mathrm{d}^3 r_s \; S(\boldsymbol{r}_s)\, \frac{\hat{\boldsymbol{n}} \cdot (\boldsymbol{r}_s - \boldsymbol{r}_w)}{|\boldsymbol{r}_s - \boldsymbol{r}_w|^3}\; K_{\mathrm{pol}}\Big(\hat{\boldsymbol{B}}(\boldsymbol{r}_s) \cdot \widehat{(\boldsymbol{r}_s - \boldsymbol{r}_w)}\Big)$$

$$K_{\mathrm{pol}}(\cos\theta) = \tfrac{3}{4}\, a\, \sin^2\theta + \Big(\tfrac{2}{3} b + \tfrac{1}{3} c\Big)\Big(\tfrac{1}{4} + \tfrac{3}{4}\cos^2\theta\Big)$$

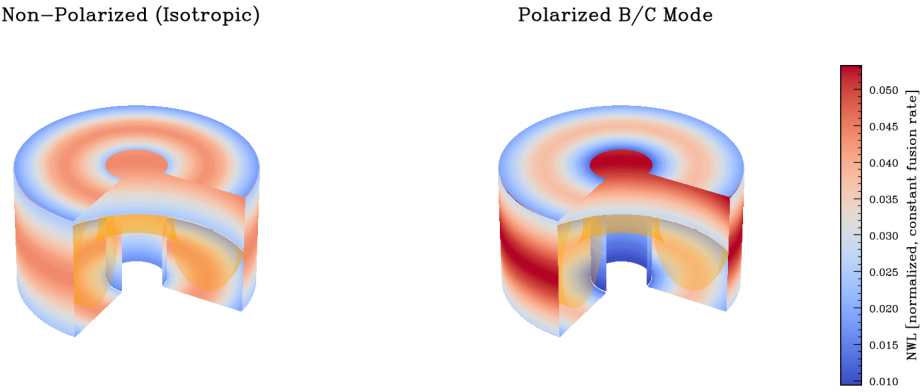
Takeaway: no scattering — a purely geometric, line-of-sight quantity, so it can be computed two independent ways and cross-checked.

OpenMC source reproduces the analytic free-streaming load



Takeaway: scattering off, the polarized MC source reproduces the analytic kernel per wall and mode (residual mean $+0.02$, σ 0.69) — sampler and line-of-sight geometry validated.

The steering, at a glance: inboard center-stack relief



Takeaway: schematic box-torus render of the computed free-streaming NWL (not CAD/DAGMC); the B/C mode moves load off the inboard center stack, the A mode does the reverse.

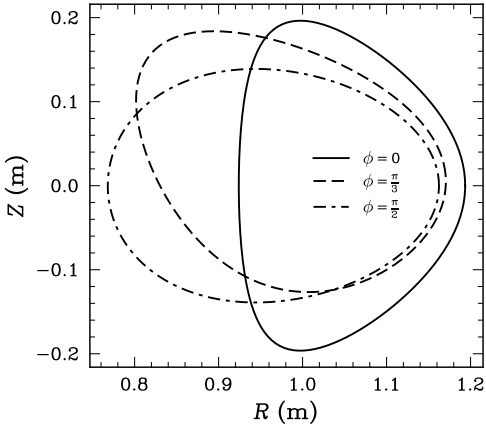
The hard part: one toroidal integral per source loop

Reduced intensity from a single source loop (the free-streaming kernel):

$$g = \int_{\phi_-}^{\phi_+} \mathcal{J}(\phi)\, \mathcal{N}(\phi)\, \frac{K\big(\cos\theta(\phi)\big)}{D(\phi)^{3/2}}\, \mathrm{d}\phi, \qquad D \equiv |\boldsymbol{\Delta}|^2$$

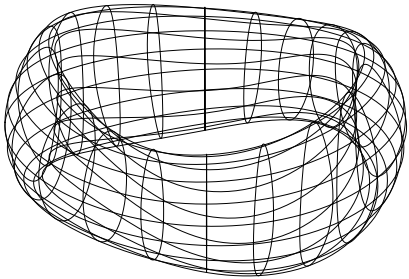
In a tokamak the loop is a planar circle $\Rightarrow$ this is elliptic (closed form). In a stellarator the loop is *non-planar* $\Rightarrow$ the integral becomes *hyperelliptic*: no finite closed form in F, E. That is the obstruction — of algebraic geometry, not of effort.

Why: the cross section rotates and reshapes with ϕ



Takeaway: the QA boundary ($N_{\mathrm{fp}} = 2$) at three toroidal angles over a half field period. This field-period excursion is exactly what the paper expands order by order to restore closed form.

Each source loop is non-planar



Takeaway: unlike a tokamak ring, every constant-(ρ, ϑ) loop carries field-period harmonics — the feature that makes the toroidal NWL integral hyperelliptic.

The idea: expand in the non-axisymmetric excursion

Write each loop as its toroidal mean plus ripple and collect order by order:

$$g = g^{(0)} + \varepsilon\, g^{(1)} + \varepsilon^2\, g^{(2)} + \cdots$$

with the leading term the axisymmetric kernel *verbatim*,

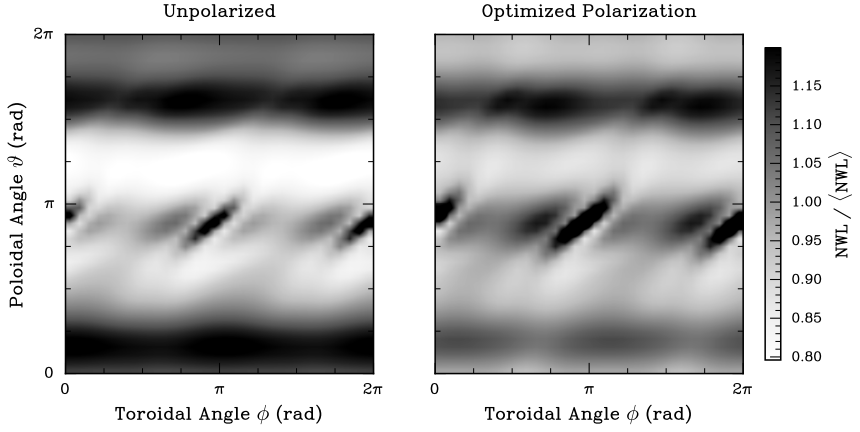
$$g^{(0)} = \int_{\phi_-^{(0)}}^{\phi_+^{(0)}} p\,\mathcal{N}_0\, \frac{K_0}{D_0^{3/2}}\,\mathrm{d}\phi$$

and every correction closed form in the *same* elliptic integrals F, E:

$$C_k^{(s)} = \int_{\phi_-}^{\phi_+} \frac{\cos k\phi}{(h-\cos\phi)^s}\,\mathrm{d}\phi \quad \xrightarrow{\text{recursions}} \quad J_{-1/2} = \tfrac{2}{\sqrt{h-1}}\big[F(\phi_+/2, m) - F(\phi_-/2, m)\big]$$

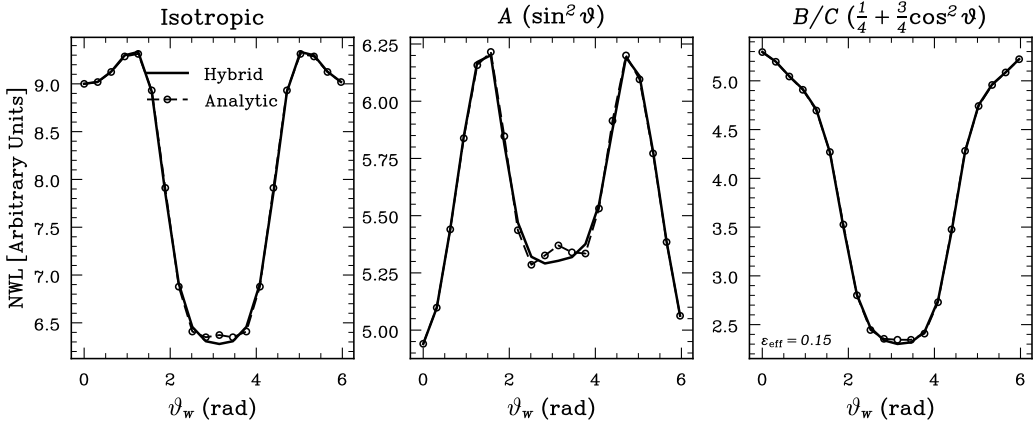
Takeaway: anchored on Schwartz's exact axisymmetric result; corrections are additive, differentiable ripple terms.

The deliverable: a full 3D free-streaming NWL map



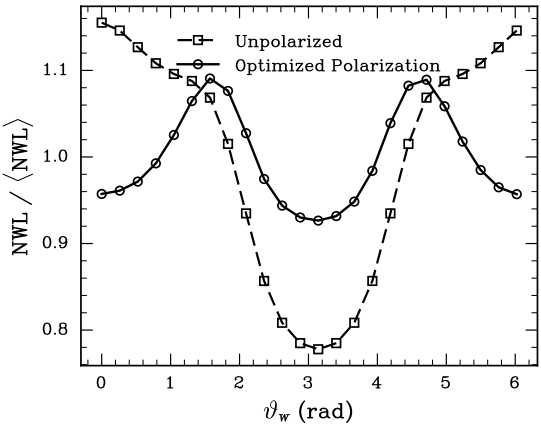
Takeaway: full toroidal×poloidal analytic NWL (unpolarized vs optimized); the $N_{\mathrm{fp}} = 2$ period shows as π-banding. One global polarization flattens it only modestly — the wall *shape* is the larger lever.

Series vs. exact free-streaming, all three modes



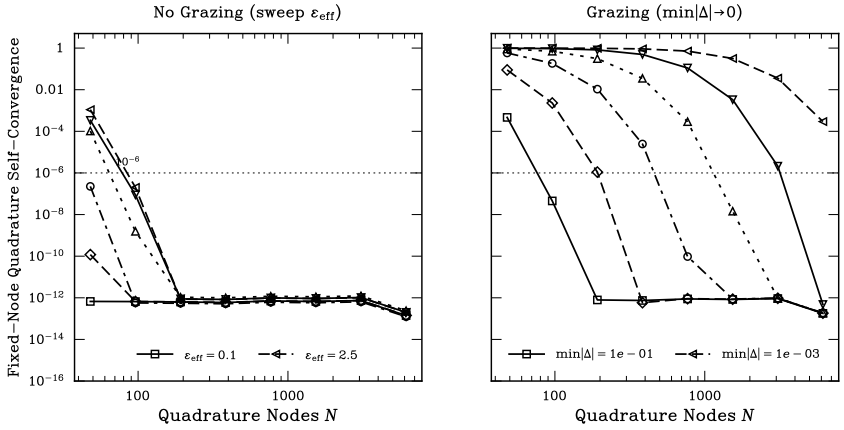
Takeaway: at $\varepsilon_{\rm eff} = 0.15$ the truncated analytic series (dashed+markers) reproduces the exact free-streaming load (solid) to $\sim$1% across isotropic, A, and B/C modes.

Differentiable $\Rightarrow$ optimizable



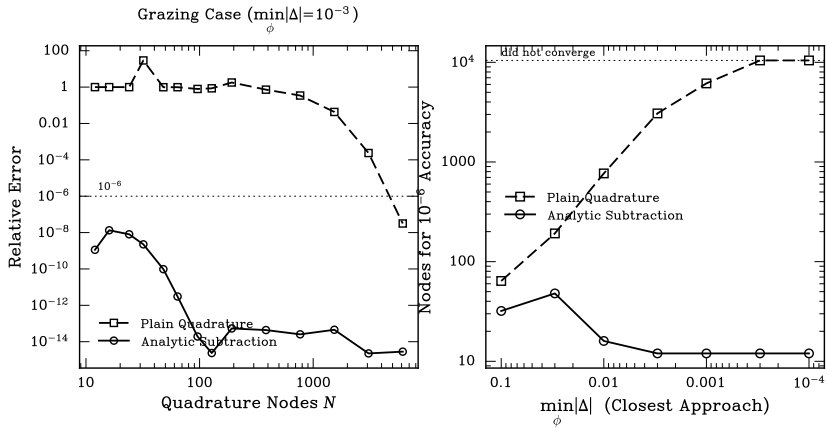
Takeaway: minimizing non-uniformity over the polarization simplex with *exact* gradients flattens the profile (squared coeff. of variation falls $\sim 6\times$) — the polarization mix is a free design knob.

When the series ends, a differentiable hybrid takes over



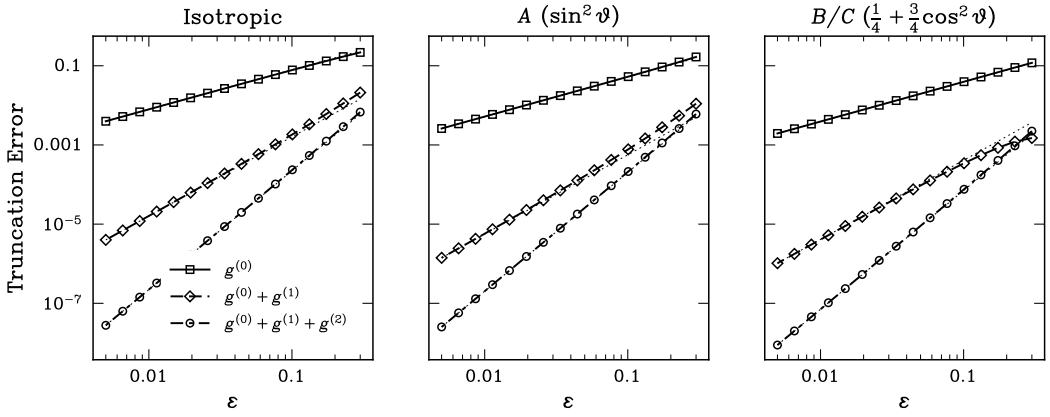
Takeaway: two thresholds, not one. The non-axisymmetry $\varepsilon_{\rm eff}$ alone does not break the quadrature; only a source filament *grazing* the wall ($\min_\phi |\mathbf{\Delta}| \to 0$) does.

Closed-form singularity subtraction handles grazing



Takeaway: the analytic structure locates the peak in closed form; subtracting it restores fixed-node convergence ($\sim$12 nodes to 10^{-6}) where plain quadrature fails.

Convergence is the predicted ε^{p+1} law



Takeaway: each retained order removes one power of the excursion amplitude (slopes $1, 2, 3$); all three polarization modes reach third order — the polarized modes only with the $|\hat{\boldsymbol{B}}| = 1$ field-normalization term.

Where this is going: HPC + DAGMC on Ginsburg

- The analytic side is the *free-streaming* ground truth. The collided wall load, the high-ε_{eff} regime, and real 3D walls need Monte Carlo transport.
- Building the full OpenMC pipeline on Columbia's **Ginsburg** cluster to get **DAGMC** — ships with OpenMC's linux-64 build as an optional feature, unavailable locally on aarch64 / Mac.
- Unlocks: a high-performance study of the *quasisymmetry effect* on the directional benefit of spin polarization across *many* configurations...
- ...including the **high-ε_{eff} regime** where the analytic series breaks down and Monte Carlo on HPC is required.

Analytic (fast, differentiable, free-streaming) $\Longleftrightarrow$ Monte Carlo (scattered, exact, HPC).